

FINA2204 Tutorial 5: Swaps

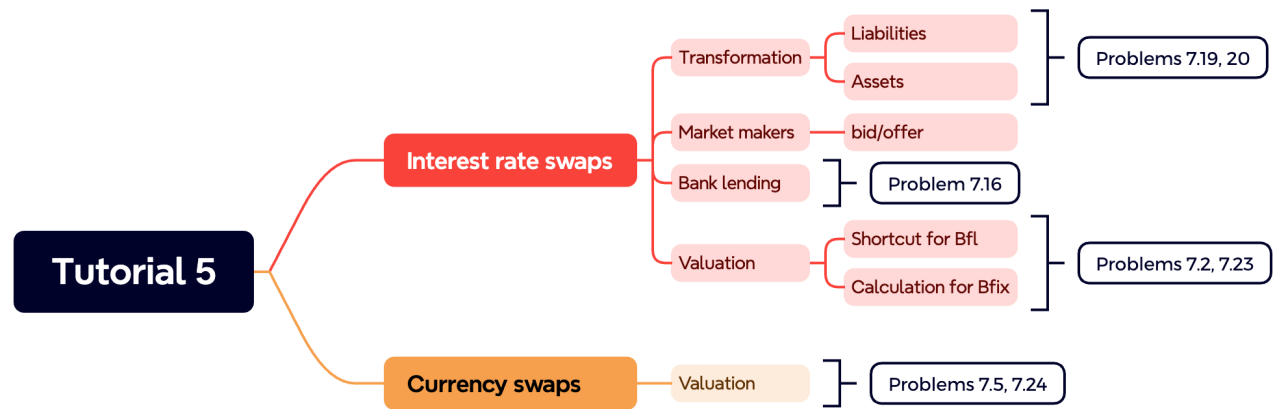
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Agenda

- Problem 7.2
- Problem 7.5
- Problem 7.16
- Problem 7.19
- Problem 7.20
- Problem 7.23
- Problem 7.24



Presented with xmind

Recap: Interest rate swaps

Key idea

A **swap** is an over-the-counter agreement between two counterparties to **exchange a stream of cash flows** in the future. The contract has to specify only two things:

1. the **dates** on which cash flows are exchanged, and
2. **how** each cash flow is calculated.

- A forward contract is essentially a swap with **only one** payment date.
- A '**plain vanilla**' interest rate swap is an exchange of a cash flow based on a **fixed** interest rate for a cash flow based on a **floating** reference rate (e.g., LIBOR, BBSW) for a predetermined number of years.

Recap: Interest rate swaps

- BBSW (Bank bill swap rates) is Australia's primary short-term benchmark interest rate.
- It represents the average cost for major banks to borrow funds from one another.

Bank Bill Swap Rates (Mid) - 10 Day History						
DATE	1 month	2 month	3 month	4 month	5 month	6 month
20/06/2024	4.2950	4.3325	4.3800	4.4988	4.5728	4.6439
19/06/2024	4.2977	4.3319	4.3768	4.4869	4.5550	4.6350
18/06/2024	4.3012	4.3321	4.3714	4.4657	4.5498	4.6090
17/06/2024	4.2950	4.3269	4.3709	4.4475	4.5300	4.5989
14/06/2024	4.2944	4.3259	4.3678	4.4400	4.5225	4.5907
13/06/2024	4.2908	4.3022	4.3659	4.4425	4.5343	4.5900
12/06/2024	4.2940	4.3067	4.3598	4.4550	4.5327	4.6135
11/06/2024	4.2914	4.3050	4.3573	4.4539	4.5250	4.6184
07/06/2024	4.2925	4.3025	4.3537	4.4296	4.5125	4.5900
06/06/2024	4.2882	4.3000	4.3458	4.4420	4.5150	4.5989


3m BBSW


6m BBSW

Recap: Interest rate swaps

Why use swaps?

1. Risk management: To switch between fixed and floating exposure as interest-rate conditions or risk preferences change.
 - *Example: Suppose a company has floating-rate debt and is worried that interest rates will rise.*
2. Flexibility: To reshape interest-rate exposure without refinancing the original loan.
 - *Example: A firm uses a swap to make the effective cost of its existing floating-rate loan fixed.*

Recap: Interest rate swaps

- Comparative advantage: To borrow where the firm has a relative cost advantage and then swap into its preferred interest-rate structure.
- *Example: A company may be able to borrow relatively cheaply in the floating-rate market even though it ultimately wants fixed-rate debt.*

Four basic cases

- **Liabilities**: floating → fixed; fixed → floating
- **Assets**: floating → fixed; fixed → floating

Recap: Transforming liabilities (case 1)

Liabilities: floating → fixed

For Apple, the swap could be used to transform a floating-rate loan into a fixed-rate loan, as indicated in Figure 7.2. Suppose that Apple has arranged to borrow \$100 million for three years at LIBOR plus 10 basis points (One basis point is 0.01%, so the rate is LIBOR plus 0.1%). After Apple has entered into the swap, it has three sets of cash flows:

floating

1. It pays LIBOR plus 0.1% to its outside lenders.
2. It receives LIBOR under the terms of the swap.
3. It pays 3% under the terms of the swap.

fixed ?

These three sets of cash flows net out to an interest rate payment of 3.1%. Thus, for Apple the swap could have the effect of transforming borrowings at a floating rate of LIBOR plus 10 basis points into borrowings at a fixed rate of 3.1%.

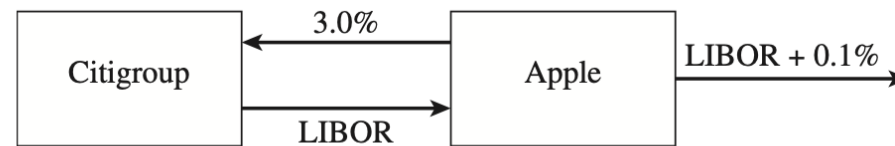


Figure 7.2 Apple uses the swap in Figure 7.1 to convert floating-rate borrowings into fixed-rate borrowings

Recap: Transforming liabilities (case 1)

	Apple
Borrowing from its outside lenders (+)	LIBOR+0.1%
Swap: Pay Citigroup (+)	3.0%
Receive from Citigroup (-)	LIBOR
Net cost of borrowing	3.1% =[(LIBOR+0.1%)+3.0%-(LIBOR)]

Recap: Transforming liabilities (case 2)

Liabilities: fixed → floating

A company wishing to transform a fixed-rate loan into a floating-rate loan would enter into the opposite swap. Suppose that Intel has borrowed \$100 million at 3.2% for three years and wishes to switch to a floating rate linked to LIBOR. Like Apple it contacts Citigroup. We assume that it agrees to enter into the swap shown in Figure 7.3. It pays LIBOR and receives 2.97%. Its position would then be as indicated Figure 7.4. It has three sets of cash flows:

1. It pays 3.2% to its outside lenders.
2. It pays LIBOR under the terms of the swap.
3. It receives 2.97% under the terms of the swap.

These three sets of cash flows net out to an interest rate payment of LIBOR plus 0.23% (or LIBOR plus 23 basis points). Thus, for Intel the swap could have the effect of transforming borrowings at a fixed rate of 3.2% into borrowings at a floating rate of LIBOR plus 23 basis points.

	Intel
Borrowing from its outside lenders (+)	3.2%
Swap: Pay Citigroup (+)	LIBOR
Receive from Citigroup (-)	2.97%
Net cost of borrowing	LIBOR+0.23% =(3.2%+LIBOR-2.97%)

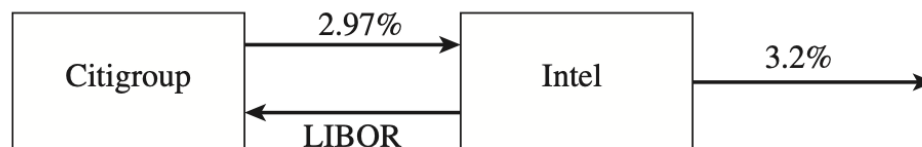


Figure 7.4 Intel uses the swap in Figure 7.3 to convert fixed-rate borrowings into floating-rate borrowings

Recap: Transforming assets (case 3)

Assets: floating $\rightarrow$ fixed

Consider next the swap entered into by Intel in Figure 7.3. The swap could have the effect of transforming an asset earning a floating rate of interest into an asset earning a fixed rate of interest. Suppose that Intel has an investment of \$100 million that yields LIBOR minus 20 basis points. After it has entered into the swap, it is in the position shown in Figure 7.6. It has three sets of cash flows:

1. It receives LIBOR minus 20 basis points on its investment.
2. It pays LIBOR under the terms of the swap.
3. It receives 2.97% under the terms of the swap.

These three sets of cash flows net out to an interest rate inflow of 2.77%. Thus, one possible use of the swap for Intel is to transform an asset earning LIBOR minus 20 basis points into an asset earning 2.77%.

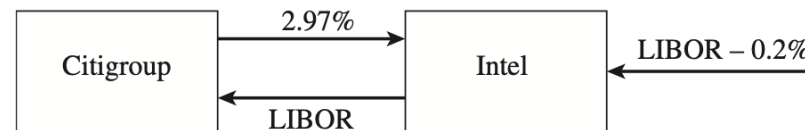


Figure 7.6 Intel uses the swap in Figure 7.3 to convert a floating-rate investment into a fixed-rate investment

Recap: Transforming assets (case 3)

	Intel
Return from its investment (+)	LIBOR-0.2%
Swap: Pay Citigroup (-)	LIBOR
Receive from Citigroup (+)	2.97%
Net return of investment	2.77% $=[(\text{LIBOR}-0.2\%)-\text{LIBOR}+2.97\%]$

Recap: Transforming assets (case 4)

Assets: fixed → floating

Swaps can also be used to transform the nature of an asset. Consider Apple in our example. The swap in Figure 7.1 could have the effect of transforming an asset earning a fixed rate of interest into an asset earning a floating rate of interest. Suppose that Apple owns \$100 million in bonds that will provide interest at 2.7% per annum over the next three years. After Apple has entered into the swap, it is in the position shown in Figure 7.5. It has three sets of cash flows:

1. It receives 2.7% on the bonds.
2. It receives LIBOR under the terms of the swap.
3. It pays 3% under the terms of the swap.

These three sets of cash flows net out to an interest rate inflow of LIBOR minus 30 basis points. The swap has therefore transformed an asset earning 2.7% into an asset earning LIBOR minus 30 basis points.

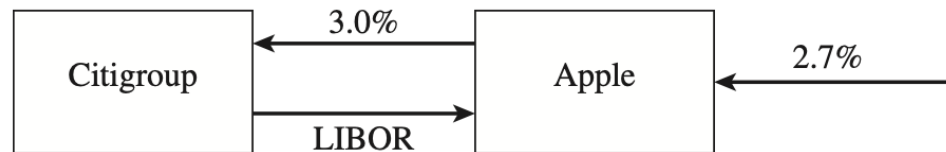


Figure 7.5 Apple uses the swap in Figure 7.1 to convert a fixed-rate investment into a floating-rate investment

Recap: Transforming assets (case 4)

	Intel
Return from its investment (+)	2.7%
Swap: Pay Citigroup (-)	3%
Receive from Citigroup (+)	LIBOR
Net return of investment	LIBOR-0.3% $=(2.7\%-3\%+\text{LIBOR})$

Recap: Market makers

- Market maker (or dealer) makers quote two **fixed rates**.
- For example, pays fixed at 2.55% to one client and receives fixed at 2.58% from another.
- Locks in a 0.03% (3 bp) **spread**.

$$+2.58\% - \cancel{LIBOR} - 2.55\% + \cancel{LIBOR} = 0.03\%$$

Two-year swap quote



Recap: Market makers

Dealers receive high and pay low on fixed rate!

- Bid: The fixed rate the dealer is willing to pay if the counterparty wants to **receive fixed** and pay floating.
- Offer (Ask): The fixed rate the dealer is willing to receive if the counterparty wants to **pay fixed** and receive floating.

Table 7.3 Bid and offer fixed rates in the swap market for a swap where payments are exchanged semiannually (percent per annum)

<i>Maturity (years)</i>	<i>Bid</i>	<i>Offer</i>	<i>Swap rate</i>
2	2.55	2.58	2.565
3	2.97	3.00	2.985
4	3.15	3.19	3.170
5	3.26	3.30	3.280
7	3.40	3.44	3.420
10	3.48	3.52	3.500

(Average bid-offer rate)

$$= \frac{\text{bid} + \text{offer}}{2}$$

Benchmark rate for **valuing swaps**, constructing swap yield curve, and as a market reference.

The actual quotes used when **entering into a swap** contract with a dealer.

Problem 7.19

(a) Company A has been offered the rates shown in Table 7.3. It can borrow for three years at 6.45%. What floating rate can it swap this fixed rate into?

Maturity (yrs)	Bid (MM)	Offer (MM)	Swap Rate
2	6.03	6.06	6.045
3	6.21	6.24	6.225
4	6.35	6.39	6.370
5	6.47	6.51	6.490
7	6.65	6.68	6.665
10	6.83	6.87	6.850

Liabilities: fixed → floating

The dealer is fixed payer, so we use the **lower** three-year quote.

Answer 7.19

- Company A can pay BBSW and receive 6.21% for three years. It can therefore transform a loan at 6.45% into a loan at BBSW plus 0.24% or BBSW plus 24 basis points.

	Company A
Borrowing from its outside lenders (+)	6.45%
Swap: Pay dealer (+)	BBSW
Receive from dealer (-)	6.21%
Net cost of borrowing	BBSW+0.24% =(6.45%+BBSW-6.21%)

Problem 7.19

(b) Company B has been offered the rates shown in Table 7.3. It can borrow for five years at BBSW plus 75 basis points. What fixed rate can it swap this floating rate into?

Maturity (yrs)	Bid (MM)	Offer (MM)	Swap Rate
2	6.03	6.06	6.045
3	6.21	6.24	6.225
4	6.35	6.39	6.370
5	6.47	6.51	6.490
7	6.65	6.68	6.665
10	6.83	6.87	6.850

Liabilities: floating → fixed

The dealer is fixed receiver, so we use the **higher** five-year quote.

Answer 7.19

- Company B can receive BBSW and pay 6.51% for five years. It can therefore transform a loan at BBSW plus 0.75% into a loan at 7.26%.

	Company B
Borrowing from its outside lenders (+)	BBSW+0.75%
Swap: Pay dealer (+)	6.51%
Receive from dealer (-)	BBSW
Net cost of borrowing	7.26% $=(\text{BBSW}+0.75\%+6.51\%-\text{BBSW})$

Problem 7.20

(a) *Company X has been offered the rates shown in Table 7.3. It can invest for four years at 5.5%. What floating rate can it swap this fixed rate into?*

Maturity (yrs)	Bid (MM)	Offer (MM)	Swap Rate
2	6.03	6.06	6.045
3	6.21	6.24	6.225
4	6.35	6.39	6.370
5	6.47	6.51	6.490
7	6.65	6.68	6.665
10	6.83	6.87	6.850

Assets: fixed → floating

The dealer is fixed receiver, so we use the **higher** four-year quote.

Answer 7.20

- Company X can pay 6.39% for four years and receive BBSW. It can therefore transform an investment at 5.5% into an investment at BBSW minus 0.89% or BBSW minus 89 basis points.

	Intel
Return from its investment (+)	5.5%
Swap: Pay Citigroup (-)	6.39%
Receive from Citigroup (+)	BBSW
Net return of investment	BBSW-0.89% =(5.5%-6.39%+BBSW)

Problem 7.20

(b) Company Y has been offered the rates shown in Table 7.3. It can invest for 10 years at BBSW minus 50 basis points. What fixed rate can it swap this floating rate into?

Maturity (yrs)	Bid (MM)	Offer (MM)	Swap Rate
2	6.03	6.06	6.045
3	6.21	6.24	6.225
4	6.35	6.39	6.370
5	6.47	6.51	6.490
7	6.65	6.68	6.665
10	6.83	6.87	6.850

Assets: floating → fixed

The dealer is fixed payer, so we use the **lower** ten-year quote.

Answer 7.20

- Company Y can receive 6.83% and pay BBSW for 10 years. It can therefore transform an investment at BBSW minus 0.5% into an investment at 6.33%.

	Intel
Return from its investment (+)	BBSW-0.5%
Swap: Pay Citigroup (-)	BBSW
Receive from Citigroup (+)	6.83%
Net return of investment	6.33% $=[(BBSW-0.5\%)-BBSW+6.83\%]$

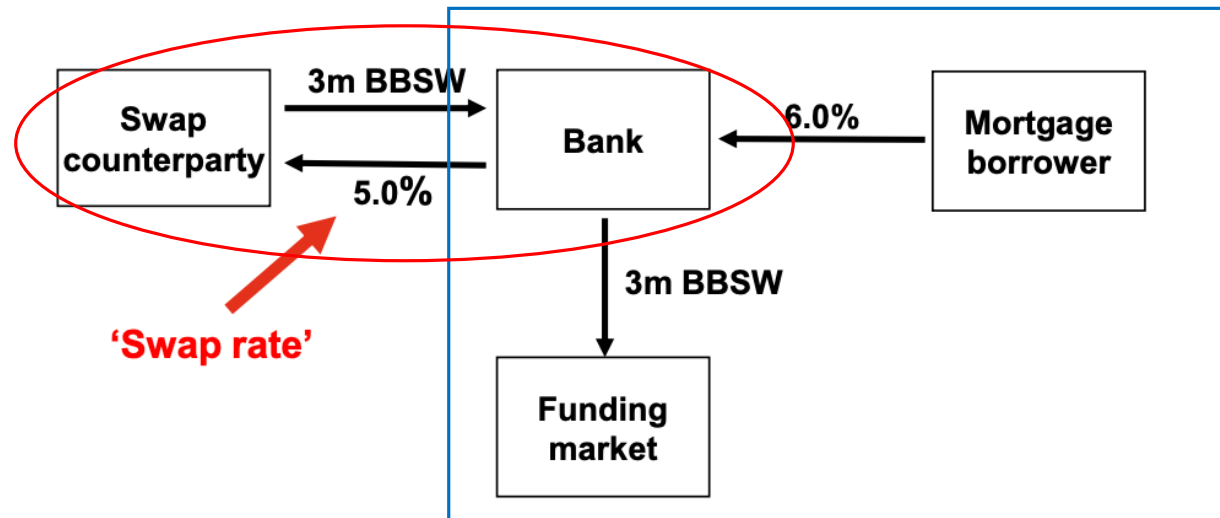
Recap: Bank lending rates

Why banks use swaps when lending fixed-rate loans?

- Banks usually fund themselves at floating rates (e.g., BBSW).
- But they lend at fixed rate (e.g., long-term fixed mortgages), their cost (floating) and income (fixed) don't match
Mismatch → Interest rate risk
- By entering a swap (pay fixed, receive floating), banks convert their floating funding into fixed, so assets and liabilities are aligned.

Recap: Bank lending rates

$$\text{Net margin} = 6\% - 3m \text{ BBSW}$$



$$\text{Net margin to bank} = 6\% - 3m \text{ BBSW} + (3m \text{ BBSW} - 5\%) = 1\%$$

$$\text{Loan rate} = \text{Swap rate} + 1\%$$

Problem 7.16

- *A bank finds that its assets are not matched with its liabilities. It is taking floating-rate deposits and making fixed-rate loans. How can swaps be used to offset the risk?*

Answer 7.16

- The bank is paying a floating-rate on the deposits and receiving a fixed-rate on the loans.
- It can offset its risk by entering into interest rate swaps (with other financial institutions or corporations) in which it contracts to **pay fixed and receive floating**.

Recap: Valuation of interest rate swaps

Bond-portfolio approach

- An interest rate swap exchanges (i) a stream of fixed cash flows for (ii) a stream of floating cash flows.
- For valuation, imagine adding the **same** notional principal to both sides at maturity.
- The added principals cancel out, so the value of the swap is unchanged.
- The two cash-flow streams can now be viewed as:
 - A fixed-rate bond, B_{fix}
 - A floating-rate bond, B_{fl}

Swap valuation becomes bond valuation!

Recap: Valuation of interest rate swaps

Key idea

To the party **receiving fixed**:

$$V_{\text{swap}} = B_{\text{fix}} - B_{\text{fl}}$$

where B_{fix} is the value of a bond paying the swap's fixed leg, and B_{fl} is the value of a floating-rate bond on the same notional. (The fixed payer holds the negative of this.)

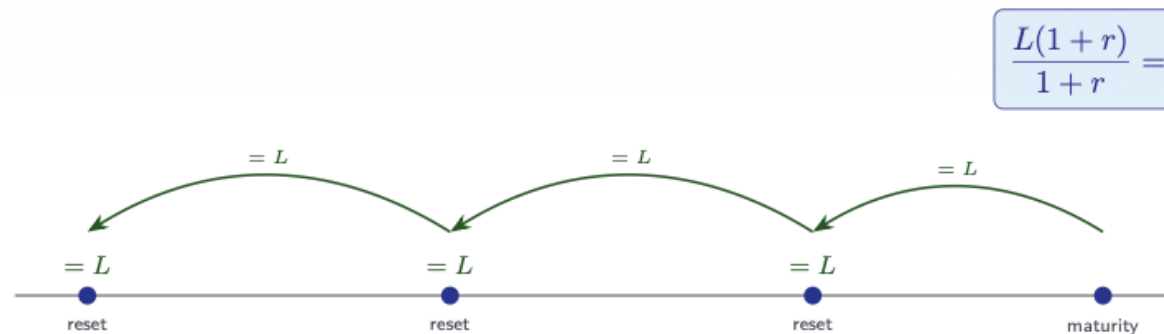
- For the party paying fixed:

$$V_{\text{swap}} = B_{\text{fl}} - B_{\text{fix}}$$

- Because a swap is a contract between two parties, the values must always be **equal and opposite**.

Recap: Valuation of interest rate swaps

Shortcut for B_{fl} : **The floating-rate resets to par!**



$$\frac{L(1+r)}{1+r} = L$$

$$\text{between resets: } B_{fl} = (L + k^*) e^{-r^* t^*}$$

1. At each reset date, immediately after the payment, the floating-rate bond is worth par:

$$B_{fl} = L$$

2. Between reset dates, the next floating payment (k^*) is already known because it was fixed at the previous reset date.

3. Therefore, discount the **next known payment (k^*) + par value (L)** back to today.

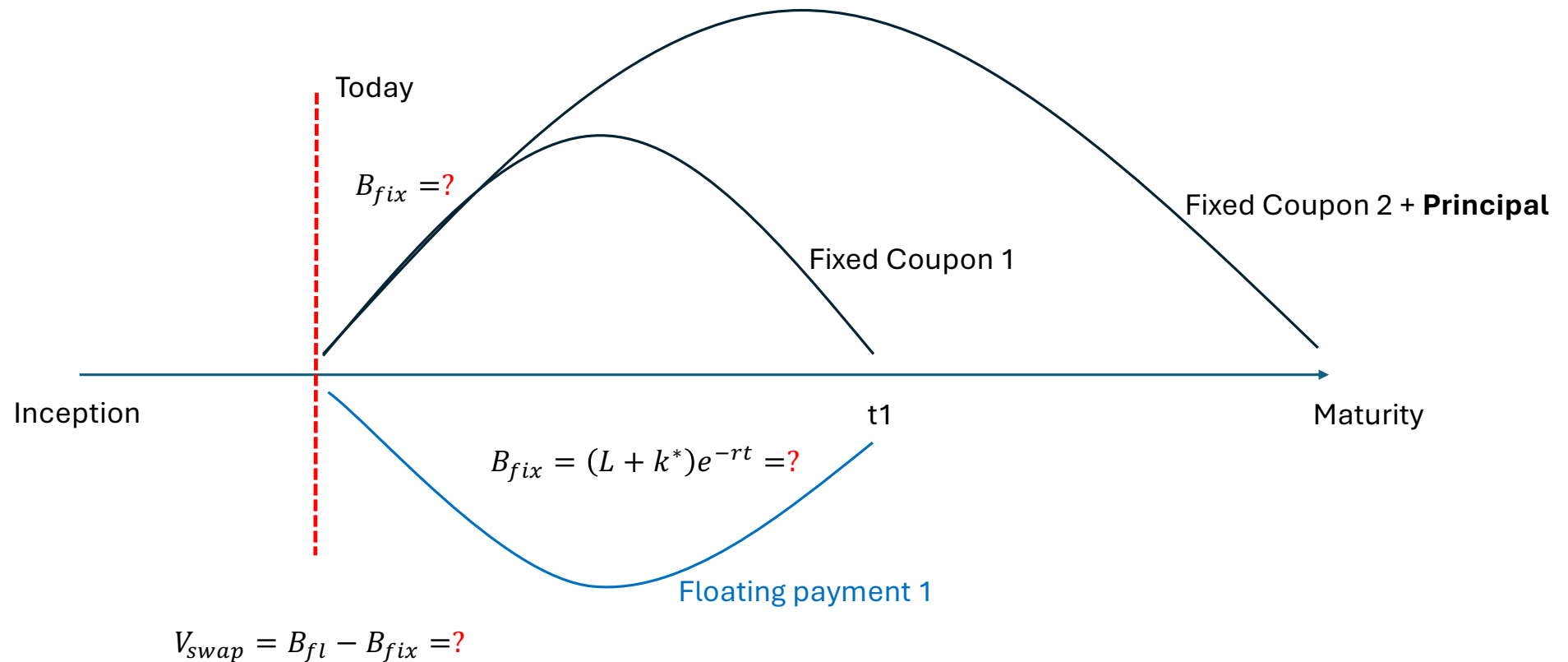
Recap: Valuation of interest rate swaps

Calculation for B_{fix}

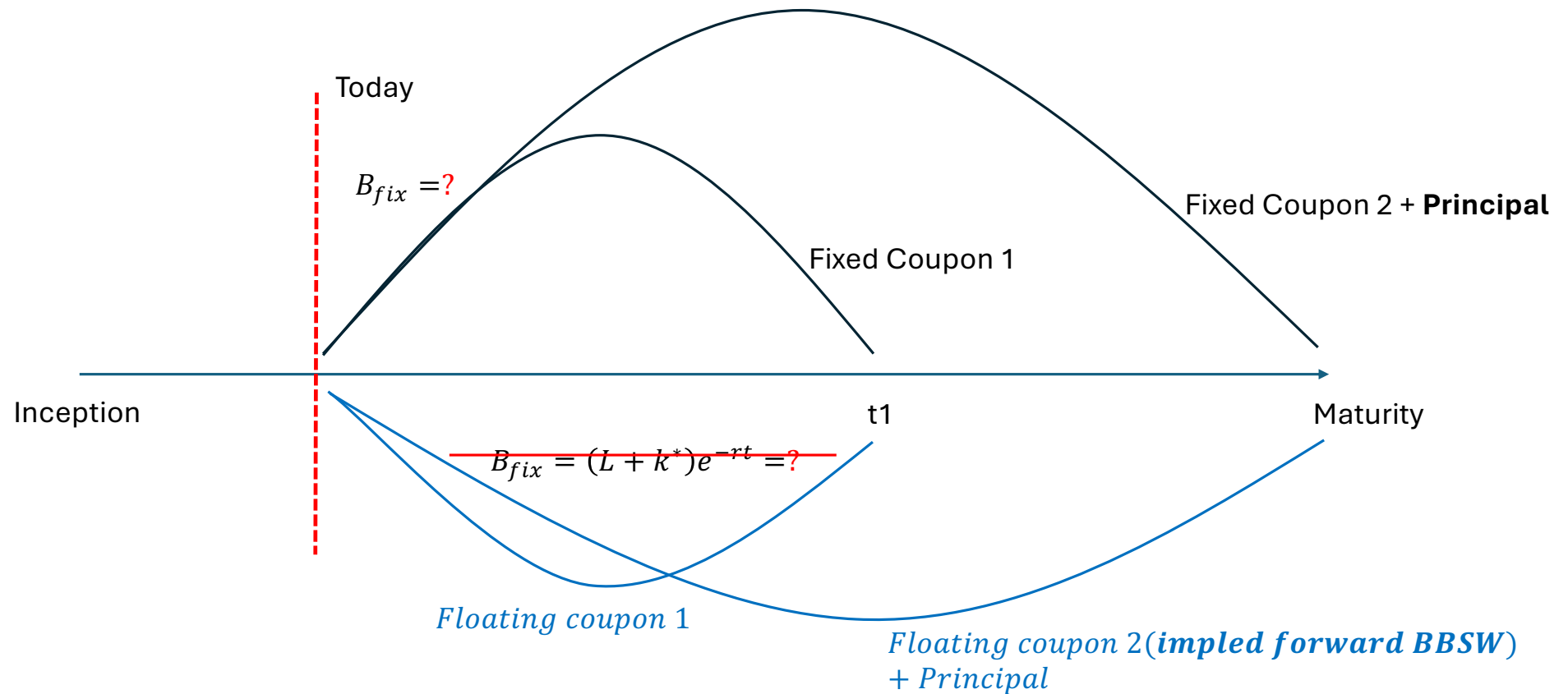
- All remaining payments are known in advance.
- Unlike the floating-rate bond, the fixed rate-bond does not reset to par ($B_{fl} \neq L$).
- Therefore, there is no reset-to-par shortcut.
- We have to discount every remaining fixed coupon **individually**, and at maturity we also include the principal.

$$B_{fix} = PV \text{ of all remaining fixed coupons} + PV \text{ of principal}$$

Recap: Valuation of interest rate swaps



Recap: Valuation of interest rate swaps



Recap: Valuation of interest rate swaps

x% per annum vs. x% (compounded annually)

- “x% per annum” is the **nominal** interest rate.
- “How often is that interest calculated?” → **compounding frequency** (annually, quarterly, or continuously)
- Discrete compounded

$$PV = FV \left(1 + \frac{R_m}{m}\right)^{-mT}$$

- Continued compounded (c.c)

$$PV = FV e^{-R_c T}$$

$$\left. \begin{array}{l} PV = FV \left(1 + \frac{R_m}{m}\right)^{-mT} \\ PV = FV e^{-R_c T} \end{array} \right\} x\% (c.c) \leftrightarrow y\% (discrete)$$

Problem 7.2

- *A \$100 million interest rate swap has a remaining life of 10 months. Under the terms of the swap, six-month BBSW is exchanged for 7% per annum (compounded semi-annually). The average of the bid-offer rates being exchanged for six-month BBSW in swaps of all maturities is currently 5% per annum with continuous compounding. The six-monthly BBSW rate was 4.6% per annum two months ago.*
- *What is the current value of the swap to the party paying floating? What is its value to the party paying fixed?*

Problem 7.2

Setup

- Notional: \$100m
- Floating (6-month BBSW) >> fixed (7%)
- Last BBSW reset rate (2 month ago): 4.6%
- Discount rate: 5% (c.c)
- Remaining life: 10 months
- Payment frequency: every 6 months
- Payment dates: month 10, month (10-6)
- $V_{swap} = ? \rightarrow B_{fix} = ? ; B_{fl} = ?$

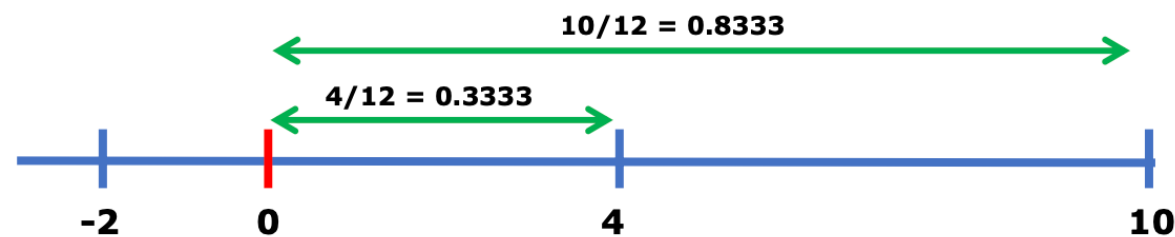
$$\rightarrow PV = FV e^{-5\%T}$$

Discount factor with continuous compounding

Answer 7.2

Floating rate shortcut version

$$\text{between resets: } B_{fl} = (L + k^*) e^{-r^* t^*}$$



Time	B_{fix} cash flow (1)	B_{fl} cash flow (2)	Discount factor (3)	PV of B_{fix} (1)*(3)	PV of B_{fl} (2)*(3)
0.333	$100 \times 7\% \times 0.5$	$100 + 100 \times 4.6\% \times 0.5$	$e^{-0.05 \times 0.333}$	$(100 \times 7\% \times 0.5) \times e^{-0.05 \times 0.333}$	$(100 + 100 \times 4.6\% \times 0.5) \times e^{-0.05 \times 0.333}$
0.833	$100 + 100 \times 7\% \times 0.5$	N/A	$e^{-0.05 \times 0.833}$	$(100 + 100 \times 7\% \times 0.5) \times e^{-0.05 \times 0.833}$	
Total				102.718	100.609

Answer 7.2

- zero curve (**flat 5%**) → Implied 6 – month forward BBSW → $(1 + \frac{R_m}{m})^{mT} = e^{R_c T} \rightarrow 1 + \frac{R_2}{2} = \frac{e^{0.05 \times 0.833}}{e^{0.05 \times 0.333}} = e^{0.05 \times 0.5} \rightarrow R_2 = 5.063\%$

5% (c.c) ↔ **5.063%** (s.a)

Time	B_{fix} cash flow (1)	B_{fl} cash flow (2)	Discount factor (3)	PV of B_{fix} (1)*(3)	PV of B_{fl} (2)*(3)
0.333	$100 \times 7\% \times 0.5$	$100 \times 4.6\% \times 0.5$	$e^{-0.05 \times 0.333}$	$(100 \times 7\% \times 0.5) \times e^{-0.05 \times 0.333}$	$(100 \times 4.6\% \times 0.5) \times e^{-0.05 \times 0.333}$
0.833	$100 + 100 \times 7\% \times 0.5$	$100 + 100 \times 5.063\% \times 0.5$	$e^{-0.05 \times 0.833}$	$(100 + 100 \times 7\% \times 0.5) \times e^{-0.05 \times 0.833}$	$(100 + 100 \times 5.063\% \times 0.5) \times e^{-0.05 \times 0.833}$
Total				102.718	100.609

Answer 7.2

- The value of the swap to the party **paying floating** is

$$V_{\text{swap}} = B_{\text{fix}} - B_{\text{fl}}$$

$$V_{\text{swap}} = 102.718 - 100.609 = \$2.109m$$

- The value of the swap to the party **paying fixed** is

$$V_{\text{swap}} = B_{\text{fl}} - B_{\text{fix}}$$
$$V_{\text{swap}} = -(102.718 - 100.609) = \$ - 2.109m$$

Problem 7.23

- *In an interest rate swap, a financial institution pays 10% per annum and receives three-month BBSW in return on a notional principal of \$100 million with payments being exchanged every three months. The swap has a remaining life of 14 months. The average of the bid and offer fixed rates currently being swapped for three-month BBSW is 12% per annum for all maturities. The three-month BBSW rate one month ago was 11.8% per annum. All rates are compounded quarterly.*
- *What is the value of the swap?*

Answer 7.23

Setup

- Notional: \$100m
- Fixed (10%) >> Floating (three-month BBSW)
- Last BBSW reset rate (1 month ago): 11.8%
- Discount rate: 12% (quarterly compounded!) $\rightarrow PV = FV \left(1 + \frac{12\%}{4}\right)^{-4T}$
- Remaining life: 14 months
- Payment dates: month 14, month (14-3), month (14-3-3), month (14-3-3-3), month (14-3-3-3-3).

Discount factor with discrete compounding

Time	B_{fix} cash flow (1)	B_{fl} cash flow (2)	Discount factor (3)	PV of B_{fix} (1)*(3)	PV of B_{fl} (2)*(3)
2/12	$100 \times 10\% \times 0.25$	$100 + 100 \times 11.8\% \times 0.25$	$(1 + 3\%)^{-4 \times 2/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 2/12}$	$(100 + 100 \times 11.8\% \times 0.25) \times (1 + 3\%)^{-4 \times 2/12}$
5/12	$100 \times 10\% \times 0.25$	N/A	$(1 + 3\%)^{-4 \times 5/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 5/12}$	
8/12	$100 \times 10\% \times 0.25$		$(1 + 3\%)^{-4 \times 8/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 8/12}$	
11/12	$100 \times 10\% \times 0.25$		$(1 + 3\%)^{-4 \times 11/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 11/12}$	
14/12	$100 + 100 \times 10\% \times 0.25$		$(1 + 3\%)^{-4 \times 14/12}$	$(100 + 100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 14/12}$	
Total				98.678	100.941

Discrete compounding: $PV = FV(1 + \frac{R_m}{m})^{-mT} \rightarrow m = 4$, discount factor = $(1 + \frac{12\%}{4})^{-4T}$

Time	B_{fix} cash flow (1)	B_{fl} cash flow (2)	Discount factor (3)	PV of B_{fix} (1)*(3)	PV of B_{fl} (2)*(3)
2/12	$100 \times 10\% \times 0.25$	$100 \times 11.8\% \times 0.25$	$(1 + 3\%)^{-4 \times 2/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 2/12}$	$(100 \times 11.8\% \times 0.25) \times (1 + 3\%)^{-4 \times 2/12}$
5/12	$100 \times 10\% \times 0.25$	$100 \times 12\% \times 0.25$	$(1 + 3\%)^{-4 \times 5/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 5/12}$	$(100 \times 12\% \times 0.25) \times (1 + 3\%)^{-4 \times 5/12}$
8/12	$100 \times 10\% \times 0.25$	$100 \times 12\% \times 0.25$	$(1 + 3\%)^{-4 \times 8/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 8/12}$	$(100 \times 12\% \times 0.25) \times (1 + 3\%)^{-4 \times 8/12}$
11/12	$100 \times 10\% \times 0.25$	$100 \times 12\% \times 0.25$	$(1 + 3\%)^{-4 \times 11/12}$	$(100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 11/12}$	$(100 \times 12\% \times 0.25) \times (1 + 3\%)^{-4 \times 11/12}$
14/12	$100 + 100 \times 10\% \times 0.25$	$100 + 100 \times 12\% \times 0.25$	$(1 + 3\%)^{-4 \times 14/12}$	$(100 + 100 \times 10\% \times 0.25) \times (1 + 3\%)^{-4 \times 14/12}$	$(100 + 100 \times 12\% \times 0.25) \times (1 + 3\%)^{-4 \times 14/12}$
Total				98.678	100.941

Implied 3 – month forward BBSW = 12%

Answer 7.23

- The value of the swap **to the party paying fixed** is therefore

$$V_{\text{swap}} = B_{\text{fl}} - B_{\text{fix}}$$
$$V_{\text{buyer}} = 100.941 - 98.678 = \$2.263m$$

Recap: Valuation of currency swaps

- Unlike interest rate swap, the fixed-for-fixed currency swap exchanges cash flows in **different currencies**;
- In an interest rate swap, the principal is only a notional amount and is not exchanged;
- In a currency swap, a **separate** principal amount is specified for each currency and the principals are typically exchanged at inception and re-exchanged at maturity.

Currency Swap

$$\underbrace{B_{GBP}^{fixed}} \leftrightarrow \underbrace{B_{AUD}^{fixed}}$$

GBP AUD

Different currencies

28/8/2026 → discount separately + FX conversion

Interest Rate Swap

$$\underbrace{B_{fix}} \leftrightarrow \underbrace{B_{fl}}$$

AUD AUD

Same currencies → compared directly

Recap: Valuation of currency swaps

- For example, imagine an Australian company wants to build a factory in the UK. They need British Pounds to pay the construction workers. However, because they are an Australian company, they can get a much cheaper loan from an Australian bank in AUD.
- They will borrow the AUD 30m, enter a currency swap to physically exchange that AUD 30m for GBP 20m today, use the GBP 20m to build their factory, and then swap the principals back at the end of the contract.

Recap: Valuation of currency swaps

Valuation in terms of a bond

1. We literally have one stream of cash flows in one currency and another stream of cash flows in another currency, including principal repayment.
2. Treat the cash flows in each currency as a bond;
3. Discount each bond using the interest rate for that currency;
4. Convert one bond value into the other currency using the spot exchange rate.

Recap: Valuation of currency swaps

- The value of a swap where domestic currency is received and a foreign currency is paid is:

$$V_{swap} = B_{domestic} - S_0 B_{foreign}$$

- Similarly, the value of a swap where the foreign currency is received and domestic currency is paid is:

$$V_{swap} = S_0 B_{foreign} - B_{domestic}$$

where S_0 represents domestic currency per unit of foreign currency.

Problem 7.5

- *A currency swap has a remaining life of 15 months. It involves exchanging interest at 10% on GBP20 million for interest at 6% on AUD30 million once a year. The term structure of interest rates in both the United Kingdom and Australia is currently flat and if the swap were negotiated today the interest rates would be 4% in dollars and 7% in sterling. All interest rates are quoted with annual compounding. The current exchange rate (dollars per pound sterling) is 1.8500.*
- *What is the value of the swap to the party paying sterling? What is the value of the swap to the party paying dollars?*

Answer 7.5

Setup

- Principals: GBP20m ↔ AUD30m
- Fixed rate: GBP 10% ↔ AUD 6%
- Discount rates: GBP 7%, AUD 4% (annually compounded!)
- Payment frequency: Annual
- Remaining life: 15 months
- Payment dates: month 15, month (15-12)

Discount factor with discrete compounding


$$\rightarrow PV = FV \left(1 + \frac{R_m}{1}\right)^{-T}$$

Answer 7.5

Discount each currency separately!

Time	B_{GBP} cash flow (1)	B_{AUD} cash flow (2)	GBP Discount factor (3)	AUD Discount factor (4)	PV of B_{GBP} (1)*(3)	PV of B_{AUD} (2)*(4)
3/12	$20 \times 10\% \times 1$	$30 \times 6\% \times 1$	$(1 + 7\%)^{-3/12}$	$(1 + 4\%)^{-3/12}$	$(20 \times 10\%) \times (1 + 7\%)^{-3/12}$	$(30 \times 6\%) \times (1 + 4\%)^{-3/12}$
15/12	$20 + (20 \times 10\%)$	$30 + 30 \times 6\%$	$(1 + 7\%)^{-15/12}$	$(1 + 4\%)^{-15/12}$	$(20 + (20 \times 10\%)) \times (1 + 7\%)^{-15/12}$	$(30 + 30 \times 6\%) \times (1 + 4\%)^{-15/12}$
Total					£22.182m	A\$32.061m

$$£1 = A\$1.85$$

GBP: Foreign currency

AUD: Domestic currency

$$GBP \text{ payer: } V_{swap} = B_{domestic} - S_0 B_{foreign}$$

$$= 32.061 - 22.182 \times 1.85 = -\$8.976m$$

$$AUD \text{ payer: } V_{swap} = S_0 B_{foreign} - B_{domestic} = \$8.976m$$

Problem 7.24

- *For all maturities, the US dollar (USD) interest rate is 7% per annum and the Australian dollar (AUD) rate is 9% per annum. The current value of the AUD is 0.62 USD. In a swap agreement, a financial institution pays 8% per annum in AUD and receives 4% per annum in USD. The principals in the two currencies are USD 12 million and AUD 20 million. Payments are exchanged every year, with one exchange having just taken place. The swap will last two more years.*
- *What is the value of the swap to the financial institution? Assume all interest rates are continuously compounded.*

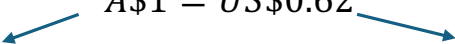
Answer 7.24

Setup

- Principals: USD12m ↔ AUD20m
- Fixed rates: USD 4% ↔ AUD 8%
- Discount rates: GBP 7%, AUD 9% (c.c)
- Payment frequency: Annual
- Remaining life: 24 months
- Payment dates: month 24, month (24-12)

Answer 7.24

Time	B_{USD} cash flow (1)	B_{AUD} cash flow (2)	USD Discount factor (3)	AUD Discount factor (4)	PV of B_{USD} (1)*(3)	PV of B_{AUD} (2)*(4)
1	$12 \times 4\% \times 1$	$20 \times 8\% \times 1$	$e^{-0.07}$	$e^{-0.09}$	$(12 \times 4\% \times 1) \times e^{-0.07}$	$(20 \times 8\% \times 1) \times e^{-0.09}$
2	$12 + (12 \times 4\%)$	$20 + 20 \times 8\%$	$e^{-0.07 \times 2}$	$e^{-0.09 \times 2}$	$(12 + (12 \times 4\%)) \times e^{-0.07 \times 2}$	$(20 + 20 \times 8\%) \times e^{-0.09 \times 2}$
Total					US\$11.297	A\$19.504

$A\$1 = US\0.62


$$\begin{aligned}
 \text{AUD payer: } V_{\text{swap}} &= B_{\text{domestic}} - S_0 B_{\text{foreign}} \\
 &= 11.297 - 0.62 \times 19.504 = -US\$0.795m
 \end{aligned}$$